

SHOPcottage

CONTAINERIZED HOBBYIST WORKSHOP SYSTEM

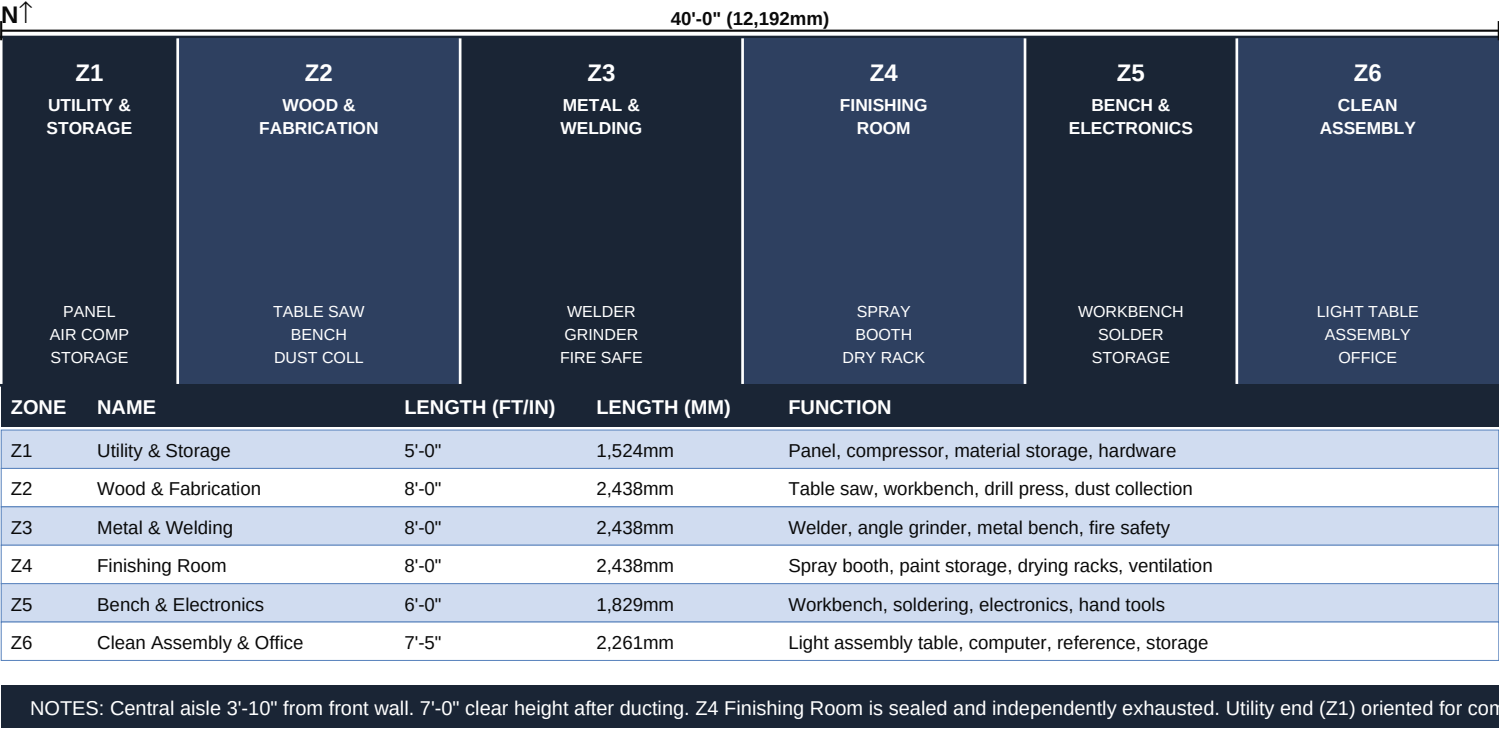
CONTAINER	WORKSHOP TYPE	CAPACITY	INTEGRATION	INSTALL TYPE
40FT HIGH CUBE 9'6" INTERIOR HT	ALL-PURPOSE HOBBYIST	1-2 OPERATORS SIMULTANEOUS	GRUSH PROJECT TRACKING	SITE BUILT ZONE 9B

This document is the official build specification for the SHOPcottage containerized hobbyist workshop system. It contains dimensioned floor plans, zone specifications, electrical single-line diagram, equipment and parts list, utility and ventilation targets, and a phased build checklist. This plan package is licensed for single-use construction. Redistribution or resale requires a contractor license from Grush.

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01 — DIMENSIONED FLOOR PLAN

40ft High Cube container interior: 39'5" × 7'8" × 8'10" usable (11,989mm × 2,337mm × 2,692mm). All dimensions to finished wall face after insulation and paneling. Zones separated by removable partition systems unless noted as fixed.



02 — ZONE SPECIFICATIONS

Z#	ZONE NAME	DESCRIPTION & KEY EQUIPMENT	PARTITION
Z1	Utility & Storage	200A main sub-panel (feeds full shop). 80-gal upright air compressor on dedicated circuit. Wall-mounted tool storage — pegboard and shelving. Consumables storage (fasteners, adhesives, abrasives). Fire extinguisher station x2. First aid cabinet. Shop-grade utility sink.	Fixed wall (plywood + FRP)
Z2	Wood & Fabrication	Contractor table saw with outfeed table. Miter saw station (wall-mount, 12in). Drill press (floor model). Router table. Full-length workbench (8ft, hardwood top). Wall-mount lumber rack. Dust collector (2HP, 4in ports). Cyclone pre-separator.	Open to aisle
Z3	Metal & Welding	MIG welder 230V (Miller or Lincoln equiv). Angle grinder x2. Metal workbench (steel top, 6ft). Bench vise (6in jaw). Cut-off saw (14in). Metal storage rack. Fire blanket + welding screen. Dedicated exhaust fan — welding fumes. Explosion-proof outlet zone.	Removable panel
Z4	Finishing Room	Spray booth enclosure — filtered intake wall + exhaust fan (explosion-proof motor). Paint and finish storage cabinet (flammable-rated). Drying rack (rolling, multi-tier). Lighting: 5000K full-spectrum for color accuracy. Positive-pressure filtered supply. No ignition sources — spark-free switches only.	Sealed panel + door sweep
Z5	Bench & Electronics	Electronics workbench (anti-static surface, 6ft). Soldering station (temp-controlled iron). Fume extractor for soldering. Oscilloscope + bench power supply. Hand tool storage — drawers and wall mount. Under-bench storage. Good overhead lighting — 5000K LED.	Open to aisle
Z6	Clean Assembly & Office	Light assembly table (laminated top, 6ft — no heavy fab). Computer workstation for plans, design, and Grush project tracking. Reference shelving (binders, parts catalogs). Parts organizer bins (wall-mount). Comfortable stool. Climate controlled — this end of container is the quietest zone.	Open to aisle

SAFETY NOTE: Z3 (Welding) and Z4 (Finishing) are the highest-hazard zones. Never operate both simultaneously without dedicated ventilation. Z4 uses explosion-proof fixtures and switches only. No open flames, sparks, or arc welding within 20ft of active spray operations.

03 — ELECTRICAL SINGLE-LINE DIAGRAM

200A service minimum (shop loads require heavy service). All branch circuits self-performed by licensed wireman. Utility tie-in by licensed electrician with permit. Cont



XP = Explosion-Proof. All Z4 Finishing Room switches, and outlets must be UL listed Class I Div 2 or equivalent. No standard switches in Z4.

CIRCUIT	VOLTAGE	EST. LOAD	ZONE	NOTES
Air Compressor	30A / 240V	~3,000W	Z1	Dedicated; hard-start capacitor recommended
MIG Welder	50A / 240V	~5,000W	Z3	Dedicated 6-50R outlet; do not share
Table Saw	30A / 240V	~2,400W	Z2	Dedicated; dust collector on same run ok
Dust Collector	20A / 120V	~900W	Z2	Interlocked with table saw if desired
Finishing Room (XP)	20A / 120V	~500W	Z4	Explosion-proof rated — all fixtures + outlets
Miter Saw / Router	20A / 120V	~1,500W	Z2	Shared circuit ok — not simultaneous
Welding Exhaust Fan	20A / 120V	~400W	Z3	Dedicated — must run during all welding ops
GFCI Outlets (x16)	20A / 120V	~1,800W	All	Duplex every 4ft along both walls
Lighting (all zones)	20A / 120V	~600W	All	5000K LED — separate circuits per half
Electronics / Computer	15A / 120V	~400W	Z5/Z6	Clean power — no heavy tool sharing
Mini-Split AC	20A / 240V	~1,500W	Z6	Comfort zone climate control

04 — EQUIPMENT & PARTS LIST

Mid-range pricing as of June 2026. Hobbyist-grade tools specified — upgrade paths noted. Prices are estimates — verify before purchase. Container and site utility costs are separate.

HVAC & CLIMATE					
ITEM	QTY	EST. UNIT \$	EST. TOTAL	ZONE	SOURCE
Mini-split AC, 12,000 BTU ductless (Z6)	1	\$750	\$750	Z6	Pioneer / MRCOOL
Shop exhaust fan, 1,200 CFM wall-mount	2	\$180	\$360	Z3/Z4	Amazon / Grainger
Explosion-proof exhaust fan (Z4)	1	\$380	\$380	Z4	Grainger
Inline fresh air fan, 6in	2	\$90	\$180	Z2/Z5	AC Infinity
Portable evaporative cooler (shop backup)	1	\$220	\$220	Z2	Amazon
ELECTRICAL					
ITEM	QTY	EST. UNIT \$	EST. TOTAL	ZONE	SOURCE
Square D QO 200A sub-panel, 40-space	1	\$320	\$320	Z1	Home Depot
GFCI duplex outlet, 20A (Leviton)	32	\$13	\$416	All	Home Depot
6-50R welder outlet + box	1	\$35	\$35	Z3	Home Depot
30A L14-30 compressor outlet + box	1	\$30	\$30	Z1	Home Depot
XP outlet, 20A, Class I Div 2 (Z4)	4	\$55	\$220	Z4	Grainger
Misc wire, conduit, boxes, hardware	lot	—	\$1,200	All	Home Depot
POWER TOOLS & MACHINES					
ITEM	QTY	EST. UNIT \$	EST. TOTAL	ZONE	SOURCE
Contractor table saw (10in, 30in rip)	1	\$600	\$600	Z2	Home Depot / Amazon
Miter saw, 12in dual-bevel	1	\$380	\$380	Z2	Amazon
Drill press, floor model, 12-speed	1	\$280	\$280	Z2	Amazon
Router table (benchtop)	1	\$250	\$250	Z2	Amazon / Rockler
Bench grinder, 8in (dual-wheel)	1	\$120	\$120	Z3	Amazon
Angle grinder, 4.5in (x2)	2	\$65	\$130	Z3	Amazon
Cut-off saw, 14in (metal)	1	\$220	\$220	Z3	Amazon
MIG welder, 230V (Lincoln 180 or equiv)	1	\$650	\$650	Z3	Amazon / Welding store
Air compressor, 80-gal upright, 175 PSI	1	\$800	\$800	Z1	Home Depot
DUST COLLECTION					
ITEM	QTY	EST. UNIT \$	EST. TOTAL	ZONE	SOURCE
Dust collector, 2HP wall-mount	1	\$380	\$380	Z2	Amazon / Woodcraft
Cyclone separator, 5-gal	1	\$120	\$120	Z2	Amazon
4in dust hose + fittings — lot	1	—	\$80	Z2	Amazon
FINISHING ROOM					
ITEM	QTY	EST. UNIT \$	EST. TOTAL	ZONE	SOURCE
HVLP spray gun, gravity feed	1	\$120	\$120	Z4	Amazon
Spray booth exhaust filter panel	1	\$180	\$180	Z4	Amazon
Flammable storage cabinet, 12-gal	1	\$220	\$220	Z4	Uline / Amazon
Rolling drying rack, 6-tier	1	\$90	\$90	Z4	Amazon
Respirator, P100 + organic vapor (x3)	3	\$40	\$120	Z4	Amazon
BENCHES & STORAGE					
ITEM	QTY	EST. UNIT \$	EST. TOTAL	ZONE	SOURCE
Heavy-duty workbench, 8ft (Z2)	1	\$350	\$350	Z2	Seville / Amazon
Steel workbench, 6ft (Z3 metal)	1	\$300	\$300	Z3	Seville / Amazon
ESD workbench mat + anti-static surface	1	\$120	\$120	Z5	Amazon
Pegboard panel, 4x8 (x4)	4	\$35	\$140	All	Home Depot
Wall-mount lumber rack	1	\$80	\$80	Z2	Amazon
Steel shelving unit, 5-shelf heavy duty	4	\$90	\$360	All	Home Depot

ITEM	QTY	EST. UNIT \$	EST. TOTAL	ZONE	SOURCE
Parts organizer bin wall board	2	\$65	\$130	Z5/Z6	Amazon
Utility sink, stainless	1	\$180	\$180	Z1	Home Depot
ELECTRONICS & FINE WORK					
ITEM	QTY	EST. UNIT \$	EST. TOTAL	ZONE	SOURCE
Soldering station, temp-controlled	1	\$80	\$80	Z5	Amazon (Hakko equiv)
Fume extractor, benchtop	1	\$80	\$80	Z5	Amazon
Bench power supply, 0-30V / 0-5A	1	\$90	\$90	Z5	Amazon
Oscilloscope, 2-channel 50MHz	1	\$280	\$280	Z5	Amazon (Rigol)
LED magnifier lamp, 5x	1	\$60	\$60	Z5	Amazon
SAFETY					
ITEM	QTY	EST. UNIT \$	EST. TOTAL	ZONE	SOURCE
Fire extinguisher, 10lb ABC (x3)	3	\$70	\$210	All	Home Depot
Fire blanket (welding area)	1	\$30	\$30	Z3	Amazon
First aid cabinet, wall-mount	1	\$80	\$80	Z1	Amazon
Safety glasses, Z87.1 (multi-pack)	1	\$40	\$40	All	Amazon
Hearing protection, NRR 30 (x4 pairs)	4	\$15	\$60	Z2/Z3	Amazon
Welding helmet, auto-darkening	1	\$120	\$120	Z3	Amazon
Smoke / CO detector (x3)	3	\$35	\$105	All	Home Depot
CONTAINER PREP					
ITEM	QTY	EST. UNIT \$	EST. TOTAL	ZONE	SOURCE
Spray foam insulation (hired crew, mat'l)	lot	—	\$800	All	Local contractor
Epoxy floor coating kit, 320 sq ft	2	\$110	\$220	All	Rust-Oleum
FRP wall panels + adhesive	lot	—	\$480	All	Home Depot
Misc hardware, fasteners, seals	lot	—	\$300	All	Home Depot

COST SUMMARY	
Total Parts (mid-range estimate)	~\$10,870
Sub labor: HVAC install + refrigerant	~\$600
Sub labor: Plumber (supply + drain)	~\$800
Sub labor: Spray foam crew	~\$900
Licensed electrician (utility tie-in only)	~\$500
Contingency (10%)	~\$1,370
TOTAL BUILD ESTIMATE (excl. container)	~\$15,040
Optional: 40HC container (used, shipyard direct) \$1,000–\$2,000	

05 — UTILITY, VENTILATION & SAFETY TARGETS

Workshop safety depends on ventilation as much as equipment. Each zone has different hazard profiles. Meet or exceed all targets before operating tools. OSHA PEL

ZONE VENTILATION & UTILITY TARGETS					
ZONE	HAZARD	VENTILATION	AIR CHANGES/HR	LIGHTING	COMPRESSED AIR
Z1 — Utility	Low	Passive + exhaust	4–6 ACH	5000K work	Yes — compressor
Z2 — Wood/Fab	Dust	Dust coll + exhaust	10–15 ACH	5000K shops	Yes — hose reel
Z3 — Welding	Fumes/UV	Dedicated exhaust fan	20–30 ACH	5000K + work	Yes — 1/4in fittings
Z4 — Finishing	Flammable	XP exhaust + filtered	30–60 ACH	5000K full-spec	Yes — HVLP spray
Z5 — Electronics	Solder fume	Fume extractor + fresh	6–10 ACH	5000K LED	Optional
Z6 — Clean Assembly	Very Low	Mini-split + passive	4–6 ACH	5000K LED	No

FIRE & SAFETY COMPLIANCE CHECKLIST

☐ Fire extinguisher (10lb ABC) within 50ft of all work zones — minimum 3 units

☐ Z4 Finishing Room: flammable storage cabinet, no ignition sources, XP rated

☐ Z3 Welding: fire blanket, clear 3ft radius, no combustibles within 35ft

☐ Smoke and CO detectors in Z1, Z3, and Z4 — interconnected if possible

☐ First aid kit mounted at Z1 entry — ANSI A+ minimum

☐ All PPE staged and accessible: glasses, ear protection, gloves, welding helmet

☐ Emergency shutoff clearly labeled at main panel — accessible without tools

☐ Exit path clear — minimum 36in aisle maintained at all times

☐ No storage within 18in of electrical panels

☐ Material Safety Data Sheets (SDS) posted for all chemical products in Z4

NOISE LEVELS — HEARING PROTECTION REQUIRED

Table Saw ~100 dB Required	MIG Welder ~80 dB Recommended	Grinder ~100 dB Required	Router ~95 dB Required	Air Comp. ~85 dB Recommended
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06 — BUILD PHASE CHECKLIST

Sequential build order. Complete each phase before beginning the next. DIY = self-performed. SUB = licensed trade required. EITHER = your call.

PHASE 1 — CONTAINER ACQUISITION & PREP	
TASK	WHO
☐ Inspect container — no fumigation residue, no structural damage	DIY
☐ Pressure wash interior, remove all debris and rust scale	DIY
☐ Apply rust treatment to any bare metal spots	DIY
☐ Install closed-cell spray foam insulation 2–3in all walls and ceiling	SUB
☐ Install 6mil vapor barrier and tape all seams	DIY
☐ Apply epoxy floor coating (320 sq ft) — 2 coats with cure time	EITHER
☐ Install FRP wall panels — Z1 through Z5 (Z6 can use painted ply)	DIY
☐ Seal all penetrations before proceeding	DIY

PHASE 2 — ELECTRICAL ROUGH-IN	
TASK	WHO
☐ Install 200A sub-panel in Z1 utility area	DIY
☐ Pull 200A service wire from site utility panel — get permit	SUB
☐ Run conduit trunk lines full length of container overhead	DIY
☐ Pull branch circuit wire to all zones per single-line diagram	DIY
☐ Install Z4 XP-rated outlets, fixtures, and switches — verify ratings	DIY
☐ Install 6-50R welder outlet in Z3 and 30A compressor outlet in Z1	DIY
☐ Install GFCI outlets every 4ft along both walls (x16)	DIY
☐ Install LED shop lights — 5000K throughout; Z6 gets highest CRI	DIY
☐ Test all circuits with load — verify breakers, GFCI trip, continuity	DIY

PHASE 3 — HVAC & VENTILATION	
TASK	WHO
☐ Install mini-split AC in Z6 — hire HVAC tech for refrigerant charge	SUB
☐ Install explosion-proof exhaust fan in Z4 (Finishing Room)	DIY
☐ Install dedicated welding exhaust fan in Z3 — verify CFM	DIY
☐ Install general shop exhaust fan in Z2 — wood dust zone	DIY
☐ Install inline fresh air fans for supply air in Z2 and Z5	DIY
☐ Install backdraft dampers on all exhaust penetrations	DIY
☐ Run all ductwork — seal all joints with mastic or foil tape	DIY
☐ Test all ventilation zones — verify target ACH by zone	DIY

PHASE 4 — PLUMBING & COMPRESSED AIR	
TASK	WHO
☐ Hire plumber — run site water supply line to container inlet	SUB
☐ Hire plumber — run drain line to site sewer or dry well	SUB
☐ Install utility sink in Z1 — connect supply and drain	SUB
☐ Position 80-gal compressor in Z1 — anchor to floor	DIY
☐ Run air supply lines — 3/4in copper or rigid to hose reel drops	EITHER
☐ Install hose reel drops in Z2 and Z3	DIY
☐ Install air dryer and regulator at compressor output	DIY
☐ Pressure test all air lines — hold 150 PSI for 30min	DIY

PHASE 5 — TOOL INSTALLATION & ZONE BUILDOUT

TASK	WHO
☐ Assemble heavy-duty workbench in Z2 — bolt to wall or floor	DIY
☐ Install table saw — align fence, set blade height, install guard	DIY
☐ Install miter saw on wall-mount station with stop blocks	DIY
☐ Install drill press — align chuck, set belt tension	DIY
☐ Mount dust collector — run 4in duct to table saw and miter saw	DIY
☐ Install cyclone separator on dust collector input	DIY
☐ Set up Z3 metal bench and vise — anchor vise to bench top	DIY
☐ Set up MIG welder — verify 6-50R outlet, test wire feed and arc	DIY
☐ Install welding screen around Z3 perimeter	DIY
☐ Build Z4 Finishing Room — filter intake wall + exhaust boot + door sweep	DIY
☐ Install flammable storage cabinet in Z4 — anchor to wall	DIY
☐ Set up Z5 electronics bench — anti-static mat, soldering station, fume extractor	DIY
☐ Set up Z6 clean assembly table and computer workstation	DIY
☐ Install pegboard, lumber rack, shelving, and parts bins throughout	DIY

PHASE 6 — SAFETY SYSTEMS & COMMISSIONING

TASK	WHO
☐ Mount fire extinguishers x3 — Z1 entry, Z2/Z3 boundary, Z4 outside	DIY
☐ Mount fire blanket at Z3 welding area	DIY
☐ Install and test smoke / CO detectors in Z1, Z3, Z4	DIY
☐ Mount first aid cabinet at Z1 entry — stock per ANSI A+ standard	DIY
☐ Label all circuits at sub-panel — permanent, legible labels	DIY
☐ Label emergency shutoff — visible from container entrance	DIY
☐ Post SDS sheets for all chemical products in Z4	DIY
☐ Verify 36in minimum aisle clearance throughout	DIY
☐ Run 48hr systems test — HVAC, ventilation, lighting, compressed air	DIY
☐ Document any deviations from plan — update as-built notes	DIY
☐ Final walkthrough and sign-off	DIY

WHO KEY: DIY = Self-performed (no labor cost) SUB = Licensed trade required EITHER = Your call / time vs money